# Sepehr Mahmoudian

## SENIOR AI ENGINEER

Machine Learning • Knowledge Graphs • Multimodal & Embodied AI




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GITHUB github.com/sepahead LINKEDIN linkedin.com/in/sepehr-mahmoudian

Berlin-Brandenburg • available from Jan 2027 • German citizen • EN fluent / DE good / Persian native


PROFILE

Builds agent harnesses for domain-specific applications, knowledge graphs and Graph RAG; orchestrates agentic workflows and evaluates LLMs, VLMs, VLAs and world models. Expertise in formal methods and verification.

LLMs / VLMs • Knowledge Graphs • Agentic Vision • Multimodal AI • Scene Reconstruction

CONTEXT ENGINEERING • DATA PROVENANCE • MODEL EVALUATION • REINFORCEMENT LEARNING

AI ENGINEERING

**10+ YEARS**

RESEARCH TO PRODUCTION

PYTHON • RUST • AI / ML

## PROFESSIONAL EXPERIENCE

### Independent AI/ML Engineer & PhD Candidate

2020.04 - Present

**Self-employed - PhD candidate at Technische Universität Darmstadt** Berlin & Darmstadt, Germany

- Building biophysical/functional neural models, PID (Partial Information Decomposition), Vision-Language-Action systems, VLA evaluation and neural control
- Building custom AI agent harnesses for domain-specific applications; multimodal 3D perception and auditable visualization

### Senior AI Engineer

2026.01 - 2026.03

**Neural Frames UG** Berlin & Hamburg, Germany

- Evaluated LLMs, VLMs and end-to-end pipelines, tracing runs and results with Langfuse
- Contributed to Python evaluation and inference pipelines; applied formal methods and verification

### AI Engineering Resident

2025.09 - 2026.01

**Merantix AI Campus (Hacker Room)** Berlin, Germany

- Built generative-media systems with Gaussian splatting and diffusion models for spatiotemporal anchoring
- Implemented confidential-computing smart contracts in Rust
- Deployed product services and data infrastructure across Cloudflare Workers, D1, R2, Google Cloud Run and AWS EC2

### Senior AI Engineer

2024.07 - 2025.07

**Vision Labs UG** Berlin & Hamburg, Germany

- Built workflows for prompt optimization, LLM evaluation and observability with DSPy and Langfuse
- Designed data and MLOps pipelines on Azure and Google Cloud with staging, automated tests and CI/CD

### Research Associate

2021.07 - 2024.07

**Freie Universität Berlin** Berlin, Germany

- Scaled and optimized large neural-network simulations in C/C++ and Python
- Built retrieval-augmented generation (RAG) workflows for technical literature and evaluated AI-agent reliability
- Applied statistical and machine-learning analysis in Python; reviewed code and research manuscripts in an international team

### Research Associate
2019.07 - 2020.06

**Georg-August-Universität Göttingen** Göttingen, Germany

Neural networks, Partial Information Decomposition (PID), information theory and reproducible data workflows

### Research Associate
2017.03 - 2019.07

**Goethe-Universität Frankfurt** Frankfurt am Main, Germany · Simulation and statistical analysis of neural-network models

### Visiting Scientist
2017.03 - 2019.03

**Max Planck Institute for Dynamics and Self-Organization** Göttingen, Germany · Statistical methods for neural-network analysis

### Research Associate
2015.02 - 2017.03

**Forschungszentrum Jülich** Jülich, Germany · Neural-network simulations and reinforcement-learning experiments

### Software Developer
2010.09 - 2012.03

**Centre for Content Creation** Cyberjaya, Malaysia · Award-winning iPhone/iPad applications and digital products

## EDUCATION

### PhD Candidate (Dr. rer. nat.)
2020.04 - Present

**Technische Universität Darmstadt** Darmstadt, Germany

*Thesis: Partial Information Decomposition and Closed-Loop Simulation of Neural and Embodied Systems on Formally Verified Foundations*

- New mathematical PID analyses and information-theoretic representation analysis; formal verification of selected results and protocol properties with Lean and SMT
- Engram's domain-specific AI agent produces neural-model candidates for closed loop neural network simulations in reproducible computational-neuroscience workflows, with provenance and bounded compilation and simulation checks

### Master's by Research in Biosciences: Neuroscience
2012 - 2014

**University College London** London, United Kingdom

Computational Neuroscience • Graduated with Distinction (Top 10%, CGPA 4.0/4.0)

### Bachelor of Science (Honours) in Computer Science
2009 - 2012

**Anglia Ruskin University** Cambridge, United Kingdom

Graduated with First Class Honours (Top 1%, CGPA 4.0/4.0)

## SELECTED AWARDS & FUNDING

**EDTH / 2025**
Special Mention at the European Defense Tech Hackathon Berlin 2025; co-organized with the Bundeswehr Cyber Innovation Hub

**HUMAN BRAIN PROJECT**
Research funding

**NEAR HORIZON**
Confidential-computing grant

**UCL & SHEFFIELD**
Competitive academic scholarships

**IOS APPLICATIONS**
15 international product awards



SIGNATURE
Sepehr Mahmoudian

PLACE Berlin, Germany
DATE 9 November 2026

PUBLIC ENGINEERING PORTFOLIO

GITHUB Open-source projects
github.com/sepahead

HUGGING FACE Models & datasets
huggingface.co/torusprime

# SELECTED PROJECTS

## engram

★ 15 · RESEARCH PROTOTYPE

**PROVENANCE-CONSTRAINED NEURAL R&D**

Domain-specific AI agent turns qualified evidence into neural-model candidates, with bounded compilation and simulation checks.

Neural Networks · Knowledge Graphs · Closed-loop Sim

## pid-rs

★ 15 · V0.9 SOURCE PRERELEASE

**VERIFIABLE INFORMATION DECOMPOSITION**

Separates redundant, unique and synergistic information, with selected Lean/SMT proofs and declared estimation assumptions.

Rust / Python · Information Theory · Lean / SMT

## prisoma

★ 5 · V0.9 PRERELEASE / ACTIVE R&D

**AUDITABLE CLOSED-LOOP EXPERIMENTS**

Simulation environment for closed-loop VLA and world-model experiments, with capture, replay and provenance.

Rust / Python · VLA + World Models · Embodied Sim

## NCP

★ 16 · SDK 1.0.1 / NCP 2 R&D

**NEURAL CONTROL PROTOCOL**

Couples simulations and tools; the next protocol combines a Lean-proved owner model with differential tests of Rust/Python SDKs.

Rust / Python · Protocol Models · Differential Tests

## manwe

★ 6 · ALPHA

**AIRSPACE PERCEPTION**

Research workbench for vision, acoustics, multi-camera geometry and tracking; model exports must pass artifact-identity and numerical gates.

Python · PyTorch · Multimodal Perception

## crebain

★ 23 · RESEARCH PROTOTYPE

**SPATIAL SITUATIONAL AWARENESS**

Unifies Gaussian-splat scenes, native object detection and multimodal 3D tracking in one tactical view for simulated drone operations.

Rust · Sensor Fusion · Native Inference

## melkor

★ 13 · HARDENING

**GAUSSIAN-SPLAT INTEGRITY**

Inspects and deterministically converts 3D Gaussian splats, surfacing numeric hazards and field provenance before pipeline use.

C++ · Scene Reconstruction · Asset Validation

## cobot-atlas
## relief-atlas

★ 8 · ↓ 3,216 · PUBLISHED + DOI

**AI-GENERATED SIMULATION DATASETS**

Cobot Atlas publishes robot-simulation meshes with a DOI; Relief Atlas develops disaster-relief assets and generation pipelines.

Python · Hugging Face · 3D / VLA

## galadriel

★ 8 · RESEARCH PROTOTYPE

**CROSS-SENSOR CONSISTENCY**

Fail-closed monitor for sensor disagreement on one track, using per-channel innovation tests and signed correlation over producer-declared projections.

Rust · NIS / CUSUM · Signed Correlation

## cortexel

★ 5 · 0.10.0-DEV.0

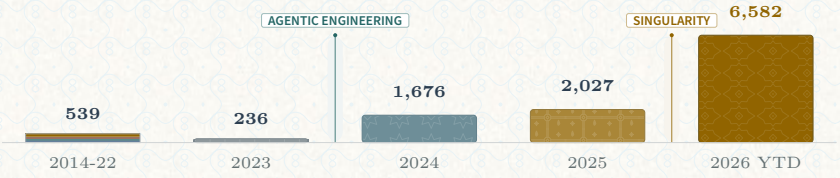
**AUDITABLE NEURAL FIGURES**

Checks scientific figures against declared contracts before rendering, so charts provably match their data and stay reproducible.

Figure Contracts · Deterministic SVG · Provenance

# PORTFOLIO & ENGINEERING ACTIVITY

| 11,060 | 23 | 125 | 200 |
|---|---|---|---|
| GITHUB CONTRIBUTIONS SINCE 2014 | NON-FORK PUBLIC REPOSITORIES | STARS ON PUBLIC REPOSITORIES | ACTIVE DAYS IN 2026 |




**CURRENT MOMENTUM**

**3.25×** 2026 YTD contribution volume versus the full 2025 calendar year

6,582 YTD • 2,027 in 2025

**CONTRIBUTION BY WEEKDAY**




6,582 contributions • 2026 YTD

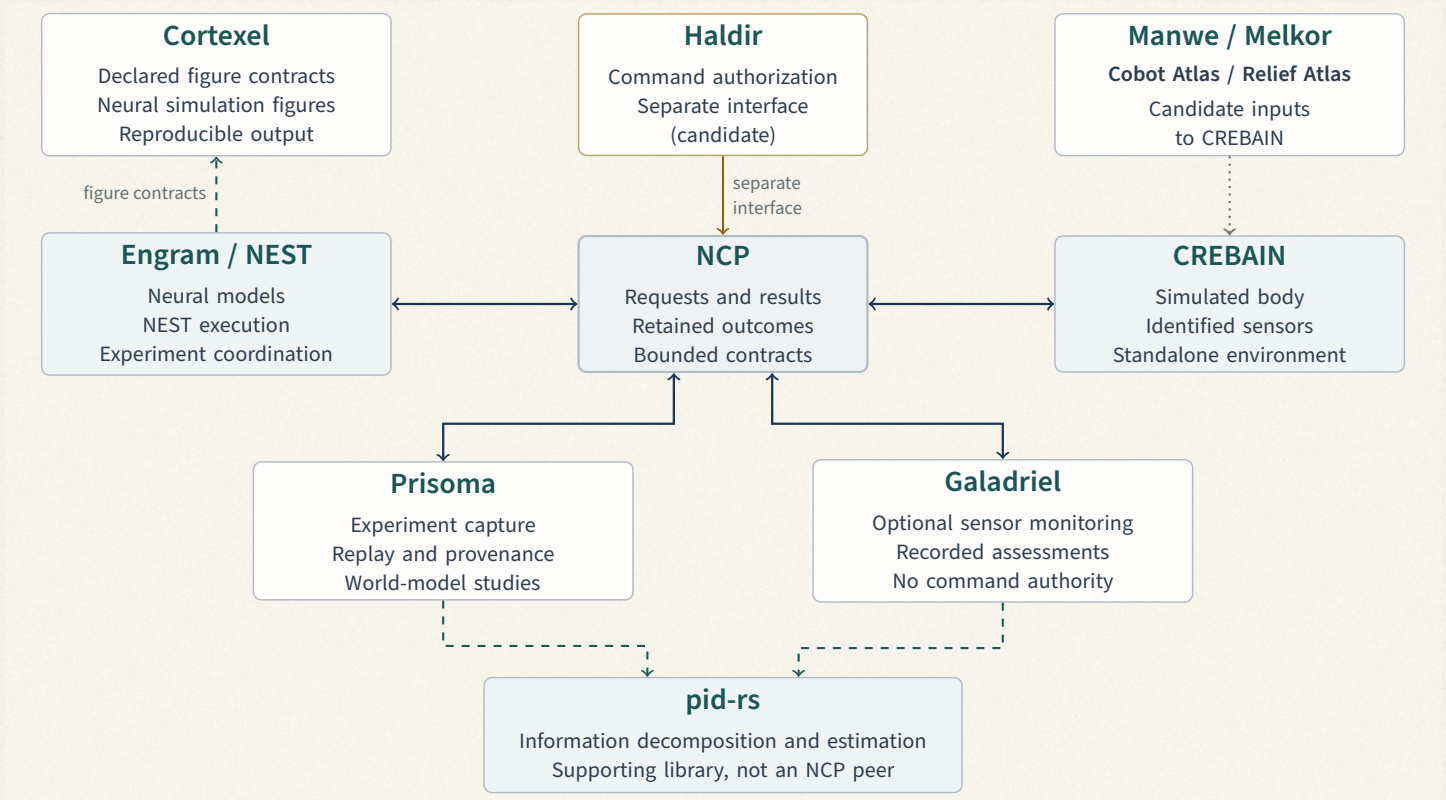
| | |
|---|---|
| Mon 17.1% | Tue 13.5% |
| Wed 12.3% | Thu 9.7% |
| Fri 10.9% | Sat 19.0% |
| Sun 17.5% | |

# TECHNICAL SKILLS

| | |
|---|---|
| AGENTIC AI | LLMs / VLMs · Agent Harnesses · AI Agents · Tool Use · Model Evaluation · Inference Pipelines |
| MACHINE LEARNING | Python · PyTorch · Agentic Vision · Multimodal AI · Vision-Language-Action / VLA Evaluation |
| KNOWLEDGE & RETRIEVAL | Knowledge Graphs · Graph RAG · RAG · Embeddings · Vector Search · Reranking · Data Provenance |
| SYSTEMS & MLOPS | Rust · C++ · FastAPI · ONNX · TensorRT · CUDA · Metal / Core ML · Docker |
| CLOUD & INFRASTRUCTURE | GCP · AWS · Azure · Kubernetes · Terraform · GitHub Actions · Linux · Zenoh |
| FORMAL VERIFICATION | Formal Methods · Lean · SMT / Z3 · Model Checking · Differential Testing |

# RESEARCH ECOSYSTEM

Independently runnable projects for neural and embodied experiments. Each program owns its state; NCP supplies shared contracts rather than a central runtime. The map distinguishes interfaces present in source from candidate inputs.




Local requests and results in source
Uses a supporting library
Candidate input; adapter not established

### Neural loop

Engram owns neural execution in NEST. CREBAIN owns the simulated body and sensor state. Selected NCP interfaces exchange action proposals and observations; the body keeps final actuation authority.

### Evidence and analysis

Prisoma captures experiment exchanges. Galadriel records optional diagnostics. Their source analysis paths use pid-rs; Engram's figure path uses Cortexel. Capture and monitoring grant no control authority.

### Qualification

Current examples start from explicit plans and scenes. Integration qualification remains partial. Manwe, Melkor and the atlases are candidate perception or asset sources. Arbitrary-paper reproduction remains a research goal.